

## Topology PhD Exam, August 2012

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Work the following problems and show all work. Support all statements to the best of your ability.

1. Prove that the 3-sphere  $S^3$  is not homeomorphic to the 3-space  $\mathbb{R}^3$ .
2. Let  $A \subset \mathbb{R}^2$  be an infinite countable subspace.
  - (a) Can  $A$  be connected?
  - (b) Is  $\mathbb{R}^2 \setminus A$  connected?
3. Construct a map  $f : T^2 \rightarrow S^2$  of degree 3 where  $T^2 = S^1 \times S^1$  is a torus.
4. Show that there is no map  $S^2 \rightarrow T^2$  of degree 3.
5. Let  $GL_n$  be the space of all invertible real  $n \times n$ -matrices. Is  $GL_n$  compact? Is it connected?

Answer the following with complete definitions or statements or short proofs.

6. Is the 2-sphere  $S^2$  with  $k$  points removed homeomorphic to a topological group for
  - (a)  $k = 1$ ?
  - (b)  $k = 2$ ?
  - (c)  $k = 3$ ?
  - (d)  $k = 0$ ?
7. State the Lefschetz Fixed Point Theorem.
8. Compute the Euler characteristic  $\chi(\mathbb{CP}^5 \times \mathbb{RP}^5 \times S^5 \times S^2)$ .
9. State the Baire Category Theorem
10. Does every function  $f : \mathbb{N} \rightarrow \mathbb{R}$  admit a continuous extension  $\bar{f} : \beta\mathbb{N} \rightarrow \mathbb{R}$  to the Stone–Čech compactification?
11. State the Five Lemma.
12. What can you say about the  $k$ -th cohomology group of a closed oriented manifold for
  - (a)  $k = n$ ?
  - (b)  $k = n - 1$ ?
13. Does there exist a covering space of the figure eight that has a non-trivial abelian fundamental group?
14. Describe all connected subsets of the real line  $\mathbb{R}$ .
15. State the Contraction Mapping Theorem.